

```
program1

import java.util.Scanner;

class areacircle
{
    int r;
    void getdata()
    {
        Scanner in=new Scanner(System.in);
        System.out.print("Enter radius:");
        r=in.nextInt();
    }
    void findarea()
    {
        double area,PI=3.14;
        area=PI*r*r;
        System.out.println("Area of circle is:"+area);
    }
}

class java1
{
    public static void main(String args[])
    {
        areacircle c1=new areacircle();
        c1.getdata();
        c1.findarea();
    }
}
```

```
}  
  
program2  
  
import java.util.Scanner;  
  
class factorial  
{  
    int fact=1,no,i;  
    void getdata()  
    {  
        Scanner in=new Scanner(System.in);  
        System.out.print("Enter no for find factorial:");  
        no=in.nextInt();  
    }  
    void findarea()  
    {  
        for(i=1;i<=no;i++)  
            fact=fact*i;  
        System.out.println("Factorial of given no is:"+fact);  
    }  
}  
  
class java2  
{  
    public static void main(String args[])  
    {  
        factorial f1=new factorial();  
        f1.getdata();  
        f1.findarea();  
    }  
}
```

```
}  
  
}  
  
program3  
import java.util.Scanner;  
class java3  
{  
    public static void main(String args[])  
    {  
        int no1,no2;  
        Scanner in=new Scanner(System.in);  
  
        System.out.println("Enter no1:");  
        no1=in.nextInt();  
  
        System.out.println("Enter no2:");  
        no2=in.nextInt();  
        if(no1>no2)  
        {  
            System.out.println("no1 is larger");  
        }  
        else  
        {  
            System.out.println("no2 is larger");  
        }  
    }  
}
```

```
}  
  
import java.util.Scanner;  
  
class java4  
{  
  
    public static void main(String args[])  
    {  
  
        int a,b,c;  
  
  
        Scanner in=new Scanner(System.in);  
  
  
        System.out.println("Enter value of a:");  
        a=in.nextInt();  
  
  
        System.out.println("Enter value of b:");  
        b=in.nextInt();  
  
  
        System.out.println("Enter value of c:");  
        c=in.nextInt();  
        if(a>b)  
        {  
            if(a>c)  
            {  
                System.out.println("value of a is larger");  
            }  
            else  
            {
```

```
                System.out.println("value of c is larger");
            }
        }
    else
    {
        if(b>c)
        {
            System.out.println("value of b is larger");
        }
        else
        {
            System.out.println("value of c is larger");
        }
    }
}

}

}

program5
import java.util.Scanner;

class java5
{
    public static void main(String args[])
    {
        int choice;

        Scanner in=new Scanner(System.in);

        System.out.println("Enter choice:");
```

```
        choice=in.nextInt();

        switch(choice)
        {

            case 1: System.out.println("This is chocie 1");
                    break;

            case 2: System.out.println("This is chocie 2");
                    break;

            case 3: System.out.println("This is chocie 3");
                    break;

            case 4: System.out.println("This is chocie 4");
                    break;

            case 5: System.out.println("This is chocie 5");
                    break;

            default: System.out.println("Please enter right choice");

        }

    }

}

program6

import java.util.Scanner;

class java6
```

```
{  
  
    public static void main(String args[])  
    {  
  
        Scanner in=new Scanner(System.in);  
  
        int no,i,a,sum=0;  
  
        System.out.println("Enter the no:");  
        no=in.nextInt();  
  
        while(no!=0)  
        {  
  
            a=no%10;  
  
            no=no/10;  
  
            sum=sum+a;  
  
        }  
  
        System.out.println("Sum of the digit is:"+sum);  
  
    }  
  
}  
  
program7  
import java.util.Scanner;  
class java7  
{  
  
    public static void main(String args[])  
    {
```

```
int no,i,a=1;

float sum=0;

Scanner in=new Scanner(System.in);


System.out.println("Enter no:");
no=in.nextInt();


for(i=1;i<=no;i++)
{
    System.out.print(i+"*"+a+"/");
    sum=sum+1.0f/i;
}

System.out.println("Sum is:"+sum);
}
}

program8

import java.util.Scanner;

class java8
{
    public static void main(String args[])
    {
        int no,i,flag=0;

        Scanner in=new Scanner(System.in);

        System.out.println("Enetr the no:");
        no=in.nextInt();
```



```
        for(i=2;i<=no/2;i++)
        {
            if(no%i==0)
            {
                flag=1;
                break;
            }
        }

        if(flag==0)
        {
            System.out.print("no is prime");
        }
        else
        {
            System.out.print("no is not prime");
        }
    }
}

program9
import java.util.Scanner;

class java9
{
    public static void main(String args[])
    {
        int no,i;
```

```
Scanner in=new Scanner(System.in);  
System.out.println("Enetr the no:");  
no=in.nextInt();
```

```
for(i=1;i<=no;i++)  
{  
    if(i==5)  
    {  
        break;  
    }  
    System.out.println(i);  
}
```

```
}
```

```
}
```

program10

```
import java.util.Scanner;
```

```
class java10
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        int no,i;
```

```
        Scanner in=new Scanner(System.in);
```

```
        System.out.println("Enetr the no:");
```

```
        no=in.nextInt();
```

```
        for(i=1;i<=no;i++)
```

```
        {
            if(i==11)
            {
                continue;
            }
            System.out.println(i);
        }
    }
}

program11
class java11
{
    public static void main(String args[])
    {
        for(String s:args)
        {
            System.out.println(s);
        }
    }
}

program12
import java.util.Scanner;

class java12
{
    public static void main(String args[])
    {
```

```
Scanner in=new Scanner(System.in);

int i,j,element,temp,no;

int arr[]=new int[10];

System.out.println("Enter the number of an array:- ");

no=in.nextInt();

for(i=1;i<=no;i++)

{

    System.out.println("Enter the element:");

    arr[i]=in.nextInt();

}

for(i=1;i<=no;i++)

{

    for(j=i;j<=no;j++)

    {

        if(arr[i] > arr[j])

        {

            temp=arr[i];

            arr[i]=arr[j];

            arr[j]=temp;

        }

    }

}

System.out.println("Sorted list in ascending order:");
```

```
        for(i=1;i<=no;i++)
        {
            System.out.println(arr[i]);
        }
    }
}

program13
import java.util.Scanner;

class Student
{
    int java,se,ec,net,t;
    float per;
    String result,grd;
    void getmarks()
    {
        Scanner in=new Scanner(System.in);

        System.out.println("Enter the java marks:");
        java=in.nextInt();

        System.out.println("Enter the software engineering marks:");
        se=in.nextInt();

        System.out.println("Enter the e-commerce marks:");
        ec=in.nextInt();
```

```
        System.out.println("Enter the net framework marks:");
        net=in.nextInt();
    }
    void total()
    {
        t=java+se+ec+net;
        System.out.println("From Total 400 get:"+t);
    }

    void percentage()
    {
        per=t/4;
        System.out.println("Percentage:"+per);
    }

    void result()
    {
        if(java<35 || se<35 || ec<35 || net<35)
        {
            System.out.println(result="Fail");
        }
        else
        {
            System.out.println(result="Pass");
        }
    }
}
```

```
    }

    void grade()
    {
        if(result=="Pass")
        {
            if(per>=70.0 || per<=100.0)
            {
                System.out.println("Student Grade:A");
            }
            else if(per>=60.0 || per<70.0)
            {
                System.out.println("Student Grade:B");
            }
            else if(per>=50.0 || per<60.0)
            {
                System.out.println("Student Grade:C");
            }
            else if(per>=40.0 || per<50.0)
            {
                System.out.println("Student Grade:D");
            }
            else if(per>=35.0 || per<40.0)
            {
                System.out.println("Student Grade:E");
            }
        }
    }
}
```

```
        }
        else
        {
            System.out.println("Student Grade:E");
        }
    }
}

class java13
{
    public static void main(String args[])
    {
        Student s1=new Student();
        s1.getmarks();
        s1.total();
        s1.percentage();
        s1.result();
        s1.grade();
    }
}
```

program14

```
import java.util.Scanner;

class empsalary
{
```



```
String name;

int id,salary,t,p,h;

Scanner in=new Scanner(System.in);

void heading()
{
    String s="Employee Salary Sleep";
    System.out.println("\t\t"+s);
}

void underline()
{
    System.out.println(".....");
}

void getdata()
{
    System.out.println("Enter emp name:");
    name=in.nextLine();

    System.out.println("Enter emp ID:");
    id=in.nextInt();

    System.out.println("Enter emp salary:");
    salary=in.nextInt();
}

void pflogic()
```

```
{  
  
    int pf=5;  
    p=salary*pf/100;  
    System.out.println("\t\tEmployee salary pf is:"+p);  
}  
  
void tddalogic()  
{  
  
    int td=10;  
    t=salary*td/100;  
    System.out.println("\t\tEmployee salary tdda is:"+t);  
}  
  
void hralogic()  
{  
  
    int hr=12;  
    h=salary*hr/100;  
    System.out.println("\t\tEmployee salary hra is:"+h);  
}  
  
void totalsalary()  
{  
  
    int total;  
    total=salary-p+t+h;  
    System.out.println("The total salary of employee is:"+total);  
}
```

```
void alldisplay()
{
    System.out.println("\t\tEmployee id:"+id);
    System.out.println("\t\tEmployee name:"+name);
    System.out.println("\t\tEmployee salary:"+salary);
}
void nesting()
{
    underline();
    heading();
    underline();
    getdata();
    heading();
    underline();
    alldisplay();
    pflogic();
    tddalogic();
    hralogic();
    underline();
    underline();
    totalsalary();
    underline();
}
}
```

```
class java14
{
    public static void main(String args[])
    {
        empsalary emp1=new empsalary();
        empsalary emp2=new empsalary();
        empsalary emp3=new empsalary();
        emp1.nesting();
        emp2.nesting();
        emp3.nesting();
    }
}
```

program15

```
import java.util.Scanner;
class staticmembers
{
    static String sem="Semester 5";
    String name;
    int rno;
    void rno()
    {
        Scanner in=new Scanner(System.in);
        System.out.println("Enter rollno:-");
        rno=in.nextInt();
    }
}
```

```
}  
  
class java15  
{  
  
    public static void main(String args[])  
    {  
  
        staticmembers s1=new staticmembers();  
  
        s1.rno();  
  
        String a=staticmembers.sem;  
  
        System.out.println("Your semester of BCA is:-"+a);  
  
    }  
}
```

program16

```
import java.util.Scanner;  
  
class nesting  
{  
  
    Scanner in=new Scanner(System.in);  
  
    int a,b;  
  
    void getdata()  
    {  
  
        System.out.println("Enter first value:-");  
  
        a=in.nextInt();  
  
  
        System.out.println("Enter second value:-");  
  
        b=in.nextInt();  
  
    }  
}
```

```
void putdata()
{
    int sum;
    sum=a+b;
    System.out.println("The sum is:-"+sum);
}

void nestingmethod()
{
    getdata();
    putdata();
}
}

class java16
{
    public static void main(String args[])
    {
        nesting n1=new nesting();
        n1.nestingmethod();
    }
}

program17
class overloading
{
    int x,y;
```

```
void mul(int a)
{
    x=a;
    System.out.println("The number is:"+x);
}

void mul(int a,int b)
{
    x=a;
    y=b;
    System.out.println("The numbers are:"+a+" and "+b);
}
}

class java17
{
    public static void main(String args[])
    {
        overloading o1=new overloading();
        o1.mul(4);
        o1.mul(5,6);
    }
}
```

Program 18

class dconstructor

```
{
    dconstructor()
    {
```

```
        System.out.println("Hi!!! kds   I am default constructor");
    }
}
class java18
{
    public static void main(String args[])
    {
        dconstructor d1=new dconstructor();
    }
}
```

Program 19

```
class pconstructor
{
    pconstructor(int x,int y)
    {
        System.out.println("Hi! I am parameterized constructor");
        int a,b,c;

        a=x;
        b=y;

        c=a+b;

        System.out.println("The sum is:"+c);
    }
}
```



```
class java19
{
    public static void main(String args[])
    {
        pconstructor p1=new pconstructor(10,20);
    }
}
```

Program 20

```
class oconstructor
{
    oconstructor()
    {
        System.out.println("Hi! I am default constructor");
    }

    oconstructor(int x)
    {
        int a=x;
        int b;
        b=a*a;
        System.out.println("The return value from single arg constructor:"+b);
    }

    oconstructor(int p,int q)
```

```
{  
  
    int j,k,s;  
  
    j=p;k=q;  
  
    s=j*k;  
  
    System.out.println("The return value from double args constructor:"+s);  
  
}  
}
```

class java20

```
{  
  
    public static void main(String args[])  
    {  
  
        oconstructor o1=new oconstructor();  
  
        oconstructor o2=new oconstructor(4);  
  
        oconstructor o3=new oconstructor(4,5);  
  
    }  
}
```

Program 21

```
import java.util.Scanner;  
  
class Singleinheritance  
{  
  
    Scanner in=new Scanner(System.in);  
  
    String sname,hobby,sem;  
  
    int rno;
```

```
void student()
{
    System.out.println("Enter Student Name:");
    sname=in.nextLine();

    System.out.println("Enter Student Hoby:");
    hobby=in.nextLine();

    System.out.println("Enter Student Semester:");
    sem=in.nextLine();

    System.out.println("Enter Student Rno:");
    rno=in.nextInt();
}

}

class Sports extends Singleinheritance
{
    int sw;

    void extra()
    {

        System.out.println("Enter Student Sport weightage:");
```

```
        sw=in.nextInt();
    }
}

class java21
{
    public static void main(String args[])
    {
        Sports s1=new Sports();
        s1.student();
        s1.extra();
    }
}
```

Program 22

```
import java.util.Scanner;

class ClassA
{
    Scanner in=new Scanner(System.in);

    String sname;

    int rno;

    void getdata()
    {
        System.out.println("Enter Student Name:");
        sname=in.nextLine();
    }
}
```

```
        System.out.println("Enter Student Rno:");
        rno=in.nextInt();
    }

}

class ClassB extends ClassA
{
    int marks;

    void total()
    {
        System.out.println("Enter Student marks:");
        marks=in.nextInt();
    }
}

class ClassC extends ClassB
{
    int sw;

    void extra()
    {

        System.out.println("Enter Student Sport weightage:");
        sw=in.nextInt();
    }
}
```

```
        }  
    }  
  
class java22  
{  
    public static void main(String args[])  
    {  
        ClassC c1=new ClassC();  
        c1.getdata();  
        c1.total();  
        c1.extra();  
    }  
}
```